WORKED SOLUTION

STEP II 1997, Question 4

Pure • Remainder Theorem

> **STEP II 1997, Question 4**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/97-s2-q4

The key idea

Division by a polynomial of degree d leaves a remainder of degree less than d, and the identity dividend $=$ divisor $\times$ quotient $+$ remainder holds for *every* x. So we substitute the values of x that make the divisor vanish: the quotient disappears and only the remainder is left. Everything in this question is that one trick, used three times.

Solution

The remainder theorem

Dividing $\mathrm{p}(x)$ by the linear polynomial $x - a$ gives a quotient $\mathrm{s}(x)$ and a remainder R whose degree is less than 1, so R is a constant:

$$\mathrm{p}(x) = (x - a)\,\mathrm{s}(x) + R \qquad \text{for all } x.$$

Since this is an identity in x, we may put $x = a$. The first term on the right becomes $0 \cdot \mathrm{s}(a) = 0$, leaving $\mathrm{p}(a) = R$. So the remainder is $\mathrm{p}(a)$, as required.

Part (i)

By the remainder theorem, the given remainders say

$$\mathrm{p}(1) = 3, \qquad \mathrm{p}(2) = 1, \qquad \mathrm{p}(3) = 5.$$

Putting $x = 1, 2, 3$ into $\mathrm{p}(x) = (x-1)(x-2)(x-3)\mathrm{q}(x) + \mathrm{r}(x)$ kills the first term each time, so

$$\mathrm{r}(1) = 3, \qquad \mathrm{r}(2) = 1, \qquad \mathrm{r}(3) = 5.$$

Ask yourself We know nothing about $\mathrm{q}(x)$. Is that a problem?

No: at $x = 1, 2, 3$ the factor $(x-1)(x-2)(x-3)$ is zero, so whatever q is, it contributes nothing. Three values pin down r because it has at most three coefficients.

A convenient way to write a polynomial of degree at most two is

$$\mathrm{r}(x) = A(x-1)(x-2) + B(x-1) + C,$$

because each substitution then removes terms one at a time:

$$\begin{aligned} x = 1: &\quad C = 3, \\ x = 2: &\quad B + C = 1 \;\Rightarrow\; B = -2, \\ x = 3: &\quad 2A + 2B + C = 5 \;\Rightarrow\; 2A - 4 + 3 = 5 \;\Rightarrow\; A = 3. \end{aligned}$$

Hence

$$\mathrm{r}(x) = 3(x-1)(x-2) - 2(x-1) + 3 = 3x^2 - 11x + 11.$$

Checking: $\mathrm{r}(1) = 3 - 11 + 11 = 3$, $\mathrm{r}(2) = 12 - 22 + 11 = 1$, $\mathrm{r}(3) = 27 - 33 + 11 = 5$.

Part (ii)

(The question writes $\mathrm{P}_n(x)$ in one place; it means the same polynomial $\mathrm{P}(x)$.) By the remainder theorem we need

$$\mathrm{P}(a) = a \qquad \text{for } a = 0, 1, \ldots, n.$$

The polynomial x already does this, but it has the wrong degree. So look at $\mathrm{D}(x) = \mathrm{P}(x) - x$ instead: we need $\mathrm{D}(a) = 0$ for $a = 0, 1, \ldots, n$, so by the factor theorem (the remainder theorem with remainder 0) each of $x, x-1, \ldots, x-n$ divides $\mathrm{D}(x)$. These $n+1$ linear factors are distinct, so their product divides $\mathrm{D}(x)$, and the product has degree $n+1$.

Ask yourself How do we make $\mathrm{P}(x)$ have degree exactly $n+1$?

Take $\mathrm{D}(x)$ to be that product itself (times a non-zero constant if we like). Adding x cannot cancel its leading term x^{n+1}, because $n+1 \geq 2$. So

$$\mathrm{P}(x) = x + x(x-1)(x-2)\cdots(x-n)$$

has degree $n+1$ and $\mathrm{P}(a) = a + 0 = a$ for each integer a with $0 \leq a \leq n$. This is a polynomial of the kind required.

In fact this argument finds them all: if P has degree $n+1$ and the stated remainders, then $\mathrm{D}(x) = \mathrm{P}(x) - x$ has degree at most $n+1$ and is divisible by the product of degree $n+1$, so $\mathrm{D}(x) = k\,x(x-1)\cdots(x-n)$ for a constant k, and $k \neq 0$ for the degree to be $n+1$. The general answer is

$$\mathrm{P}(x) = x + k\,x(x-1)(x-2)\cdots(x-n), \qquad k \neq 0.$$

Checks

- For $n = 1$, $\mathrm{P}(x) = x + x(x-1) = x^2$: dividing by x leaves 0 and dividing by $x-1$ leaves 1, as needed.

- In (i), take any q, say $\mathrm{q}(x) = x$, form p and divide by $x-1$, $x-2$, $x-3$ by synthetic division: the remainders come out as $3, 1, 5$.
- We checked by computer the remainder theorem on 1000 random polynomials, part (i) with 100 random choices of q, and part (ii) for $n = 1, \ldots, 11$ with $k = 1, -3, 7$; every remainder agreed exactly.

Common slips

- Forgetting that in part (ii) the polynomial $\mathrm{P}(x) = x$ itself satisfies the conditions but has degree 1, not $n+1$; the product term is what fixes the degree.
- Solving three simultaneous equations in the coefficients of $ax^2 + bx + c$ is fine but slower; the nested form $A(x-1)(x-2) + B(x-1) + C$ avoids the elimination.

Beyond STEP

Part (i) is an instance of polynomial interpolation: through any m points with distinct x-values passes exactly one polynomial of degree less than m, and the nested form we used is Newton's form of it. The same reasoning, in the language of rings, is the Chinese remainder theorem for polynomials: knowing a polynomial modulo $x-1$, $x-2$ and $x-3$ is the same as knowing it modulo their product. Allenby's *Rings, Fields and Groups* develops polynomial rings and this theorem.

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